MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 02 (SOLUTIONS)
Chapter 3: Pair of Linear Equations in Two Variables

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (3 × 1 = 3 Marks)

Q1. If the system of equations $2x + 3y = 7$ and $(a + b)x + (2a - b)y = 21$ has infinitely many solutions, then: [1]

(A) $a = 1, b = 5$ (B) $a = 5, b = 1$ (C) $a = -1, b = 5$ (D) $a = 5, b = -1$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B)** $a = 5, b = 1$ [0.5 Mark]

We apply the condition for infinitely many solutions:

Formula / Theorem Used: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Justification: [0.5 Mark]

$\frac{2}{a+b} = \frac{3}{2a-b} = \frac{7}{21} = \frac{1}{3} \implies a + b = 6$ and $2a - b = 9$. Adding both: $3a = 15 \implies \mathbf{a = 5}$, $\mathbf{b = 1}$.

Q2. The value of k for which the system of equations $x + 2y = 3$ and $5x + ky + 7 = 0$ has no solution is: [1]

(A) 10 (B) 6 (C) 3 (D) 1

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** 10 [0.5 Mark]

Rewriting in standard form: $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$. Using the parallel-lines condition:

Formula / Theorem Used: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Justification: [0.5 Mark]

$\frac{1}{5} = \frac{2}{k} \neq \frac{-3}{7} \implies k = 2 \times 5 = \mathbf{10}$ (check $\frac{1}{5} \neq -\frac{3}{7}$).

Q3. If $am \neq bl$, then the system of equations $ax + by = c$ and $lx + my = n$: [1]

(A) has a unique solution (B) has no solution (C) has infinitely many solutions (D) may or may not have a solution

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) has a unique solution** [0.5 Mark]

We know the unique-solution determinant condition:

Formula / Theorem Used: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \iff a_1b_2 \neq a_2b_1 \implies$ Unique Solution

Justification: [0.5 Mark]

Here $a_1 = a, b_1 = b$ and $a_2 = l, b_2 = m$. Dividing $am \neq bl$ by lm (or comparing cross-products) gives $\frac{a}{l} \neq \frac{b}{m}$, so the lines intersect at a unique point.

SECTION B: Assertion-Reason Question MATHS MANTHAN | (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The pair of equations $2x + 3y = 7$ and $kx + \frac{9}{2}y = 12$ has no solution if $k = 3$.

Reason (R): The lines representing $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: At $k = 3$, $\frac{a_1}{a_2} = \frac{2}{3}$, $\frac{b_1}{b_2} = \frac{3}{9/2} = \frac{6}{9} = \frac{2}{3}$, and $\frac{c_1}{c_2} = \frac{-7}{-12} = \frac{7}{12}$. [0.5 Mark]

Since $\frac{2}{3} = \frac{2}{3} \neq \frac{7}{12}$, the lines are parallel and the system has no solution. Hence both (A) and (R) are true and (R) correctly explains (A).

SECTION C: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q5. Find the value of k for which the pair of linear equations $kx + 3y = -(k - 3)$ and $12x + ky = -k$ has infinitely many solutions. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $kx + 3y + (k - 3) = 0$ and $12x + ky + k = 0$.

Formula / Theorem Used: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \implies$ Infinitely Many Solutions

Step 1: Equating Coefficient Ratios [1 Mark]

$$\frac{k}{12} = \frac{3}{k} = \frac{k-3}{k}$$

From the first pair: $\frac{k}{12} = \frac{3}{k} \implies k^2 = 36 \implies k = \pm 6$.

Step 2: Checking the Second Equality (c_1/c_2) [1 Mark]

From $\frac{3}{k} = \frac{k-3}{k}$ (with $k \neq 0$): $3 = k - 3 \implies k = 6$.

(Verification at $k = -6$: $\frac{3}{-6} = -\frac{1}{2}$ whereas $\frac{-6-3}{-6} = \frac{3}{2} \neq -\frac{1}{2}$, so $k = -6$ is rejected.)

$$\boxed{\mathbf{k = 6}}$$

Q6. Find the values of a and b for which the following system of linear equations has infinitely many solutions:

$$2x + 3y = 7 \quad \text{and} \quad (a+b)x + (2a-b)y = 21$$

[2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Applying Ratio Condition [1 Mark]

For infinitely many solutions, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$:

$$\frac{2}{a+b} = \frac{3}{2a-b} = \frac{7}{21} = \frac{1}{3}$$

Step 2: Solving Simultaneous Equations for a and b [1 Mark]

$\frac{2}{a+b} = \frac{1}{3} \implies a + b = 6$ — (i) and $\frac{3}{2a-b} = \frac{1}{3} \implies 2a - b = 9$ — (ii)

Adding (i) and (ii): $3a = 15 \implies a = 5$. Substituting in (i): $5 + b = 6 \implies b = 1$.

$$\boxed{\mathbf{a = 5, \quad b = 1}}$$

SECTION D: Short & Long Answer Questions MATHS MANTHAN | ($1 \times 3 + 1 \times 5 = 8$ Marks)

Q7. Solve the following pair of linear equations graphically:

$$x - y + 1 = 0 \quad \text{and} \quad 3x + 2y - 12 = 0$$

Determine the coordinates of the vertices of the triangle formed by these lines and the x-axis, and shade the triangular region. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Generating Point Tables [1 Mark]

$x - y + 1 = 0 \implies y = x + 1$; $3x + 2y = 12 \implies y = \frac{12-3x}{2}$:

x	-1	0	2
$y = x + 1$	0	1	3

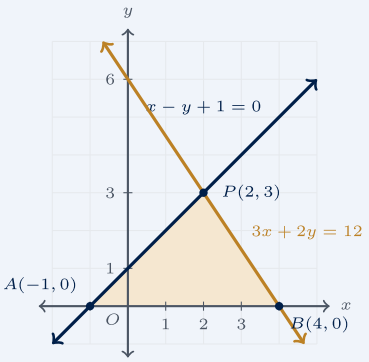
x	0	4	2
$y = \frac{12-3x}{2}$	6	0	3

Step 2: Plotting & Intersection [1 Mark]

Both lines intersect at $P(2, 3)$, giving unique solution $\mathbf{x = 2}$, $\mathbf{y = 3}$.

Step 3: Vertices of Shaded Triangle [1 Mark]

The lines intersect the x-axis ($y = 0$) at $A(-1, 0)$ and $B(4, 0)$. The shaded triangle is $\triangle PAB$ (Base $= 5$, Height $= 3$, Area $= 7.5$ sq. units).



$$\boxed{\text{Solution: } \mathbf{(2, 3)}; \quad \text{Triangle Vertices: } \mathbf{(-1, 0)}, \ \mathbf{(4, 0)}, \ \mathbf{(2, 3)}}$$

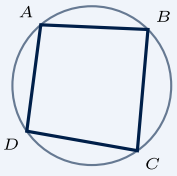
Q8. $ABCD$ is a cyclic quadrilateral. Find the measures of all four angles of the cyclic quadrilateral, given that:

$$\angle A = 4y + 20^\circ, \quad \angle B = 3y - 5^\circ, \quad \angle C = -4x^\circ, \quad \text{and} \quad \angle D = -7x + 5^\circ$$

[5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

We apply the Cyclic Quadrilateral Property:



Formula / Theorem Used: $\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$

Step 1: Setting Up Linear Equations [2 Marks]

$\angle A + \angle C = 180^\circ$: $(4y + 20) + (-4x) = 180 \implies -4x + 4y = 160 \implies x - y = -40$ — (i)

$\angle B + \angle D = 180^\circ$: $(3y - 5) + (-7x + 5) = 180 \implies -7x + 3y = 180$ — (ii)

Step 2: Solving for x and y [2 Marks]

From (i), $y = x + 40$. Substituting into (ii):

$$-7x + 3(x + 40) = 180 \implies -7x + 3x + 120 = 180 \implies -4x = 60 \implies x = -15$$

Thus $y = -15 + 40 = 25$.

Step 3: Finding All Four Angles [1 Mark]

$\angle A = 4(25) + 20 = 120^\circ$, $\angle B = 3(25) - 5 = 70^\circ$, $\angle C = -4(-15) = 60^\circ$, $\angle D = -7(-15) + 5 = 110^\circ$.

$$\boxed{\angle A = \mathbf{120^\circ}, \quad \angle B = \mathbf{70^\circ}, \quad \angle C = \mathbf{60^\circ}, \quad \angle D = \mathbf{110^\circ}}$$

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Q9. CASE STUDY: Urban Auto-Rickshaw Fare Structure

In a smart city, the passenger auto-rickshaw fare consists of a fixed hiring charge together with a distance-based charge per kilometre travelled. A student, Aditya, travelled 8 km across City Sector A and paid Rs. 91. On another day, for a continuous commute of 14 km, he paid Rs. 145. Let the fixed charge be Rs. x and the running charge per kilometre be Rs. y. [4]

(i) Formulate the pair of linear equations in x and y representing the two journeys. [1]

(ii) Find the fixed charge (x) and the running charge per kilometre (y). [1]

(iii) If a passenger travels a distance of 25 km, calculate the total fare payable. [1]

(iv) If a passenger paid an amount of Rs. 235, find the total distance travelled by him. [1]

— OR —

Find the difference between the running charge for 18 km and the fixed charge. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: Let fixed charge = Rs. x, charge per km = Rs. y.

Formula / Theorem Used: Total Fare = Fixed Charge (x) + (Distance in km) × (Rate per km y)

Sub-part (i): Formulating Equations [1 Mark]

8 km commute: $x + 8y = 91$ — (i) and 14 km commute: $x + 14y = 145$ — (ii)

Sub-part (ii): Solving for x and y [1 Mark]

Subtracting (i) from (ii): $(x + 14y) - (x + 8y) = 145 - 91 \implies 6y = 54 \implies y = 9$.

Substituting $y = 9$ into (i): $x + 8(9) = 91 \implies x = 91 - 72 = 19$.

$$\boxed{\text{Fixed charge } \mathbf{x} = \text{Rs. } \mathbf{19}, \qquad \text{Running charge } \mathbf{y} = \text{Rs. } \mathbf{9}/\text{km}}$$

Sub-part (iii): Total Fare for 25 km [1 Mark]

Fare $= x + 25y = 19 + 25(9) = 19 + 225 = \boxed{\text{Rs. } \mathbf{244}}$.

Sub-part (iv): Distance Travelled for Rs. 235 [1 Mark]

$19 + 9d = 235 \implies 9d = 235 - 19 = 216 \implies d = \frac{216}{9} = \boxed{\mathbf{24} \text{ km}}$.

Sub-part (iv) [OR Option]: Difference (18 km Running Charge − Fixed Charge) [1 Mark]

Running charge for 18 km $= 18y = 18 \times 9 =$ Rs. 162. Difference $= 162 - 19 = \boxed{\text{Rs. } \mathbf{143}}$.

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